

MARCELO VENTURINI

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PROFILE

Recent M.Sc. Biomedical Engineering graduate with a strong electronics-oriented background and hands-on experience in biomedical instrumentation, electronic measurements, embedded systems, sensor interfacing, and signal processing. Proficient with oscilloscopes, multimeters, signal generators, MATLAB/Simulink, Python, and microcontroller programming.

EDUCATION

M.Sc. in Biomedical Engineering, Politecnico di Torino, Italy 2024 – 2026
Final grade: **109/110**. Thesis: *EMG Signal Deconvolution as Preprocessing for Enhanced Hand Gesture Recognition*.

B.Sc. in Biomedical Engineering, Universidad Nacional de Córdoba, Argentina 2020 – 2024
Final grade: **8.54/10**. Awarded a competitive scholarship for international study at Politecnico di Torino.
Relevant coursework: Digital Electronics, Biomedical Instrumentation, Signal Processing, Embedded Systems.

TECHNICAL SKILLS

Test & Lab	Oscilloscopes, multimeters, signal generators, soldering, electronic measurements, hardware debugging
Programming	MATLAB, Simulink, Python, Assembler, Arduino IDE, Excel
Embedded & HW	ESP32, PIC microcontrollers, sensor interfacing, Multisim
Languages	Spanish (native), English (C1), Italian (fluent)

RELEVANT EXPERIENCE

Teaching Assistant – Digital Electronics 2024
Universidad Nacional de Córdoba

- Supported undergraduate laboratory sessions and practical exercises in digital electronics.
- Assisted students with logic circuits, measurements, debugging, and interpretation of experimental results.

SELECTED PROJECTS

Levodopa Response Assessment Device Incubated Project

- Contributed to the development of a biomedical device to objectively assess Parkinson's patients' response to levodopa treatment through sEMG acquisition, sensor integration, and signal processing.
- Supported MATLAB-based acquisition and analysis workflows to extract quantitative physiological measures for objective evaluation of treatment response.

Ultrasonic Sensor Interfacing with PIC Microcontroller Academic Project

- Implemented low-level timing, acquisition, and processing logic to interface an ultrasonic sensor with a PIC microcontroller programmed in assembler.
- Strengthened embedded debugging and measurement skills through direct hardware interfacing and timing-sensitive low-level development.

ADDITIONAL EXPERIENCE

Sailing Coach 2022–2025
Hampton YC (USA), Cold Spring Harbor (USA), Huguenot YC (USA), and Orza Minore (Italy)